INAYA NETWORK

TECHNICAL DOCUMENTATION

PROJECT TECHNICAL SCOPE OF WORK

Inaya Protocol — Technical SOW

DePIN Custody Layer: Component Deliverables, Architecture Boundaries & System Data Flow

Target Audience Technical Auditors · Venture Capital Architects · Core Engineering Teams

Document Version 1.0.4-TECH

System Class DePIN Custody Layer

Network BNB Chain (EVM Native)

Classification Public

SECTION 01

Project Executive Objective

> *A production-grade, zero-knowledge DePIN for high-value enterprise data custody — replacing centralized database clusters with a trustless pipeline.*

The technical objective of Project Inaya is to design, implement, and scale a production-grade, zero-knowledge Decentralized Physical Infrastructure Network (DePIN) for high-value enterprise data custody. The platform completely replaces centralized database clusters with a trustless pipeline consisting of:

1. Client-side browser runtime encryption and binary midpoint slicing.
2. Multi-swarm distributed content routing via the InterPlanetary File System (IPFS).
3. On-chain state attestation registries deployed via Ethereum Virtual Machine (EVM) architecture on the BNB Chain.

SECTION 02

Technology Stack Boundaries

The project execution is strictly confined to the following software development matrices and network environments:

Layer	Technologies
Client Interface & Runtime	Next.js (React), TailwindCSS, Node.js, window-injected Web3 provider APIs (e.g. MetaMask runtime API)
Cryptographic Libraries	Native Web Crypto API, PBKDF2 parameters, authenticated AES-GCM-256 bit blocks
On-Chain Ledger	Solidity `^0.8.20` , Hardhat / Foundry, OpenZeppelin standard library, BNB Chain Testnet / Mainnet
Distributed Storage Core	IPFS decentralized swarm configurations, localized API gateways, multi-tenant telemetry relays

SECTION 03

Core Component Deliverables (In-Scope)

Component A — Zero-Knowledge Client-Side Slicing Module

IN SCOPE

KEY DERIVATION ENGINE

Implementation of a PBKDF2 derivation pipeline that isolates administrative master passkeys (P) using random 16-byte salt strings (S) over 100,000 iterations via HMAC-SHA256.

BINARY MIDPOINT SLICER

Development of a dynamic string/buffer segmentation script that calculates the exact array midpoint of the encrypted payload, splitting the binary object into independent, non-contiguous streams: Shard Alpha and Shard Beta.

$$M_p = \lceil \text{length}(C) / 2 \rceil$$

TRANSPORT RELAY HANDSHAKE

Asynchronous stream upload triggers to push Shard Alpha (Vα) and Shard Beta (Vβ) concurrently to completely isolated network storage locations.

Component B — On-Chain EVM Smart Contract State Registry

IN SCOPE

ASSET RECORD SCHEMA

Design of secure, immutable Solidity structs to map custom asset identification strings against decentralized storage content identifiers (CIDs).

```
struct AssetRecord {
  string filename;
  string cidAlpha;   // CID pointing to Swarm A
  string cidBeta;    // CID pointing to Swarm B
  uint256 timestamp;
  address operator;
}
```

COLLISION AVOIDANCE LOGIC

Hardened require assertions to prevent overwriting existing file index identifiers (asset collision guard).

EVENT LOGGING PIPELINE

Implementation of indexed EVM event emitters (emit AssetArchived) to enable real-time front-end transaction scanning and interface updating.

Component C — B2B Integration SDK & High-Throughput Gateways

IN SCOPE

PLUG-AND-PLAY NPM PACKAGE

Packaging the core encryption, sharding, and contract handshake architecture into a clean, distributable package (@inaya-network/custody-sdk).

DUAL-ASSET SETTLEMENT ROUTER

Developing smart contract verification loops to enforce the hybrid pricing matrix (0.1 USDT + 0.1 INAYA per gigabyte) during execution triggers.

DEDICATED ENTERPRISE GATEWAYS

Allocation protocols for routing corporate B2B requests through dedicated RPC nodes to meet strict institutional Service Level Agreements (SLAs).

SECTION 04

System Architecture & Data Flow Protocol

The data transit flow must strictly follow the path modeled below:

User Uploads Asset

▼

Next.js Client Runtime: PBKDF2 Key Derivation → AES-GCM-256 Encryption

▼

Binary Slicing Engine: Midpoint Split $M_p = \lceil length(C)/2 \rceil$

▼

SHARD ALPHA (Vα)	SHARD BETA (Vβ)
Deployed IPFS Cluster A Generates cidAlpha	Deployed IPFS Cluster B Generates cidBeta

▼

EVM Smart Contract Handshake: Dual-Asset Fee Approval 0.1 USDT + 0.1 INAYA

▼

BNB Chain State Registry — Immutable Attestation Logged

SECTION 05

Out of Scope (Excluded Developments)

To maintain structural focus and clear timeline execution milestones, the following elements are explicitly declared out of scope for the current MVP / V1 implementation phase:

- ✕ Native physical server operating system kernels (OS development for independent nodes).
- ✕ Cross-chain token bridges outside EVM-compatible layers.
- ✕ Zero-knowledge SNARK proof generation for individual decentralized file storage verification circuits (deferred to V2.0 optimization pipelines).

"A trustless pipeline, precisely bounded — every component scoped, every boundary declared."

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